Detailed Validation of a Cartesian-to-RAMSES-Octree-to-RASCAS Pipeline against the Analytical Homogeneous Static-Sphere Lyα Solution

Reproducible workflow note

13 August 2026

Abstract

I construct a homogeneous, static, neutral-hydrogen sphere at $T = 100$ K in a 64^3 Cartesian array, convert it to the RAMSES octree binary format, create a spherical RASCAS domain of radius 0.3 box units, emit monochromatic central Lyα photon packets, and compare the escaped spectra with the analytical static-sphere solution. I test centre-to-surface optical depths $\tau_0 = 10^5, 10^6$, and 10^7. The pipeline is validated independently at the Cartesian, RAMSES, RASCAS-domain, photon-input, and escaped-photon stages. The peak locations, double-peaked morphology, normalization, and red–blue symmetry agree with theory. The normalized full-profile L_1 difference improves from 0.1124 at $\tau_0 = 10^5$ (one million photons) to 0.0649 at 10^6 and 0.0446 at 10^7 (one hundred thousand photons each), consistent with the increasing validity of the analytical diffusion approximation as $a_v\tau_0$ increases. This document explains every physical choice, conversion factor, binary interface, calculation, plot, discrepancy, and acceptance decision step by step.

Contents

1 Purpose and validation philosophy

My aim is not merely to produce a double-peaked spectrum. I want to demonstrate that the same physical problem survives the complete chain

$$\begin{aligned}&\text{physical target} \longrightarrow \text{Cartesian NumPy fields} \longrightarrow \text{RAMSES octree files}\\&\longrightarrow \text{RASCAS spherical domain} \longrightarrow \text{Monte Carlo photons} \longrightarrow \text{escaped spectrum.}\end{aligned}$$

At each arrow I perform an independent check. This is important because a plausible spectrum could still be produced with a wrong unit conversion, a reversed ionization field, an incorrect field index, a temperature error, or a diameter optical depth used where the analytical formula expects a radius optical depth.

I use three optical-depth cases. The 10^5 case is a useful diagnostic but is not deep in the asymptotic diffusion regime. The 10^6 and 10^7 cases are the stronger analytical tests. For the initial controlled comparison I use 10^5 photon packets in each optical-depth case. I additionally run 10^6 packets at $\tau_0 = 10^5$ to show which differences are Monte Carlo noise and which are systematic profile-shape differences.

2 Definition of the physical experiment

2.1 Geometry and gas state

I choose a homogeneous sphere centred at

$$\boldsymbol{x}_{\rm c} = (0.5, 0.5, 0.5)$$

inside a unit computational box. The spherical boundary radius is

$$R = 0.3\, L_{\rm box}.$$

The code length unit is one kiloparsec,

$$L_{\rm box} = \text{unit\_l} = 3.085677581282 \times 10^{21}\ \text{cm},$$

and therefore

$$R = 9.257032743846 \times 10^{20}\ \text{cm} = 0.3\ \text{kpc}.$$

I set $T = 100$ K and use primordial hydrogen mass fraction $X_{\rm H} = 0.76$. The gas has zero bulk velocity, zero turbulent velocity, zero metallicity, zero dust, and zero ionized fraction. Cosmological expansion, recoil, and helium-ion fields are absent. The source is a point at the sphere centre, with zero source velocity, emitting monochromatic photons at $\lambda_\alpha = 1215.67$ Å. Scattering is isotropic.

These choices isolate the basic resonant-transfer physics. With no preferred direction or velocity, the escaped spectrum must be symmetric about line centre. Any persistent red–blue asymmetry would be evidence of a numerical, geometrical, or interface problem.

2.2 Optical-depth convention

I define the line-centre optical depth from the centre to the spherical surface:

$$\tau_0 = n_{\rm HI}\, \sigma_0(T)\, R. \tag{1}$$

It is not the optical depth across the diameter. This convention must match the analytical solution and the RASCAS escape boundary. Using $2R$ by mistake would shift the expected peaks by $2^{1/3}$ and would be a genuine physics failure.

3 Atomic constants and thermodynamic quantities

3.1 RASCAS-compatible line quantities

I calculate the constants from the installed RASCAS atomic definition rather than inserting rounded textbook values. The adopted values are

$$m_{\rm ion} = 1.008\, m_{\rm u}, \qquad f_{12} = 0.416, \qquad A_{21} = 6.265 \times 10^8\ {\rm s^{-1}}.$$

At temperature T, the hydrogen thermal speed and Doppler width are

$$v_{\rm th} = \sqrt{\frac{2k_{\rm B}T}{m_{\rm ion}}}, \tag{2}$$

$$\Delta\nu_D = \frac{v_{\rm th}}{\lambda_\alpha}. \tag{3}$$

At 100 K I obtain

$$v_{\rm th} = 1.2844045175\ {\rm km\,s^{-1}}, \qquad \Delta\nu_D = 1.0565404406 \times 10^{10}\ {\rm Hz}.$$

The Voigt damping parameter is

$$a_{\rm v} = \frac{A_{21}}{4\pi\Delta\nu_D} = 4.7187295447 \times 10^{-3}. \tag{4}$$

The line-centre cross-section is evaluated with the same COLT approximation to $H(a_{\rm v}, 0)$ that is used by the installed RASCAS source:

$$\sigma_0 = \frac{\sqrt{\pi}e^2 f_{12}}{m_e c\,\Delta\nu_D} H(a_{\rm v}, 0) = 5.8642983367 \times 10^{-13}\ {\rm cm^2}. \tag{5}$$

Matching this convention matters because a small cross-section mismatch becomes a direct optical-depth mismatch.

3.2 Temperature in code units

The velocity and density units are

$$\text{unit\_v} = 10^5\ {\rm cm\,s^{-1}}, \qquad \text{unit\_d} = 6.7699139342702 \times 10^{-23}\ {\rm g\,cm^{-3}}.$$

For neutral primordial gas I use

$$\mu = \frac{1}{X_{\rm H} + (1 - X_{\rm H})/4} = 1.2195121951. \tag{6}$$

The RAMSES temperature conversion is

$$T = \frac{P}{\rho}\,\mu \frac{m_p}{k_B}\,\text{unit\_v}^2. \tag{7}$$

Therefore the required pressure-to-density ratio at 100 K is

$$\frac{P}{\rho} = 0.6768607801.$$

For $\gamma = 5/3$ and $P = (\gamma - 1)\rho u$, the internal energy field is

$$u = \frac{P/\rho}{\gamma - 1} = 1.0152911701. \tag{8}$$

4 Choosing the density for each optical depth

I rearrange 1 to obtain

$$n_{\mathrm{HI}} = \frac{\tau_0}{\sigma_0 R}. \tag{9}$$

The corresponding RAMSES dimensionless mass density is

$$\rho_{\mathrm{RAMSES}} = \frac{n_{\mathrm{HI}} m_p}{X_{\mathrm{H}}\,\mathrm{unit\_d}}. \tag{10}$$

4.1 Mandatory converter density multiplier

The Cartesian converter does not write its input density unchanged. It applies the built-in multiplier

$$F_\rho = \frac{h_0^2}{a_{\mathrm{exp}}^3} = \frac{0.6736^2}{0.125^3} = 232.31332352. \tag{11}$$

Although this factor has a cosmological form, I leave the converter unchanged. I compensate for it in the input array:

$$\rho_{\mathrm{Cartesian}} = \frac{\rho_{\mathrm{RAMSES}}}{F_\rho}. \tag{12}$$

If I inserted the desired RAMSES density directly into the NumPy array, the converted gas would be too dense and τ_0 would be too large by a factor of approximately 232.3.

Table 1: Densities derived without fitting or post-conversion rescaling.

τ_0	n_{HI} [cm^{-3}]	$\rho_{\mathrm{Cartesian}}$	ρ_{RAMSES}
10^5	$1.8420955095 \times 10^{-4}$	$2.5777397642 \times 10^{-8}$	$5.9884329180 \times 10^{-6}$
10^6	$1.8420955095 \times 10^{-3}$	$2.5777397642 \times 10^{-7}$	$5.9884329180 \times 10^{-5}$
10^7	$1.8420955095 \times 10^{-2}$	$2.5777397642 \times 10^{-6}$	$5.9884329180 \times 10^{-4}$

5 Step 1: constructing the Cartesian fields

For every optical depth I make C-contiguous NumPy `float64` arrays of shape $(64, 64, 64)$. I use the axis order `array[x,y,z]`, which is the order sampled by the converter. I fill the complete cube homogeneously; I do not apply a spherical mask at this stage. The sphere is imposed later as the exact RASCAS computational boundary.

The fields are

File	Assigned value
`density.npy`	case-dependent value in 1
`velocity_x.npy`, `velocity_y.npy`, `velocity_z.npy`	exactly zero
`metallicity.npy`	exactly zero
`internal_energy.npy`	1.0152911701
`scalar_01.npy`	$x_{\mathrm{HII}} = 0$ exactly
`pressure.npy`	diagnostic only; ignored by the converter

The converter derives pressure internally as

$$P = (\gamma - 1)\rho u, \tag{13}$$

so the saved pressure array is only a human-readable diagnostic. After writing each field I reload it from disk and check shape, dtype, C-contiguity, finiteness, homogeneity, exact zeros, and min/-max/mean. Central-slice images are constant, as required for a homogeneous experiment.

6 Step 2: converting the Cartesian cube to a RAMSES octree

I pass a validation-owned JSON configuration with `--config`; I do not modify the converter's original configuration. I run only the AMR and hydro writers because particles are neither present nor required by `CreateDomDump`.

The AMR metadata uses

$$\texttt{levelmin} = \texttt{levelmax} = 6, \qquad \texttt{boxlen} = 1.$$

The converter records the normal hierarchy through levels 1–6, but every physical leaf is at level 6. A level-6 uniform mesh has

$$2^6 \times 2^6 \times 2^6 = 64^3 = 262,144$$

leaf cells. This exact count is recovered by the trusted RAMSES reader and `yt` for all three cases.

The hydro field order written and later read is density, three velocity components, pressure, metallicity, internal energy, and `scalar_01`. I explicitly verify the field order because using `itemp`, `imetal`, or `ihii` with the wrong positions could produce a physically plausible but incorrect RASCAS mesh.

The converted validations recover

- exactly 262,144 level-6 leaves;
- homogeneous case-dependent density after the factor in 11;
- $u = 1.0152911701$ and the derived pressure;
- $T = 100$ K to floating-point accuracy;
- exactly zero velocity, metallicity, and `scalar_01`.

7 Step 3: creating and validating the RASCAS spherical domain

I run `CreateDomDump` with a sphere centred at $(0.5, 0.5, 0.5)$ and radius 0.3. The important reader settings are

`ramses_rt=T`, `itemp=5`, `imetal=6`, `ihii=8`,

with absent helium ion fields set to zero, `cosmo=F`, `vturb_kms=0`, and `ignoreDust=T`.

The mesh dump does not retain $x_{\rm HII}$ itself. I therefore validate $\texttt{scalar\_01} = 0$ upstream in RAMSES, then check downstream that the dumped $n_{\rm HI}$ equals the expected total hydrogen density. Each domain contains 33,696 selected level-6 leaves in 5,553 octs.

I calculate columns along the six Cartesian axes and 32 reproducibly seeded random rays. Every ray begins at the source and is clipped at the exact spherical boundary. For a uniform sphere,

$$N_{\rm HI} = \int_0^R n_{\rm HI}\, ds = n_{\rm HI} R, \qquad \tau_0^{\rm measured} = \sigma_0 N_{\rm HI}. \tag{14}$$

All 38 rays recover the requested optical depth to floating-point precision, without density fitting or rescaling. The dumped temperature is 100 K and speed, turbulence, and dust density are exactly zero.

8 Step 4: constructing and validating photon packets

I use `PhotonsFromSourceModel` to create unformatted sequential Fortran files. Every packet has

- source position $(0.5, 0.5, 0.5)$;
- source velocity $(0, 0, 0)$;
- an isotropically sampled unit direction;
- wavelength 1215.67 Å, corresponding to a single finite line-centre frequency;
- deterministic global seed -20260813 and an individual packet seed.

The photon input file contains records in this order: packet count, represented real-photon rate, global seed, packet IDs, frequencies, positions, directions, packet seeds, and source velocities. My streaming validator checks the leading and trailing Fortran record markers and supports logical records larger than 2 GiB. It confirms the exact packet count, sequential unique IDs, central positions, zero velocities, monochromatic finite frequencies, and unit direction norms.

9 Step 5: RASCAS transport assumptions

RASCAS reads the validated spherical domain and photon file. I use isotropic HI scattering, no recoil, no dust, no bulk velocity, and no turbulent velocity. I set `nDirections=0`; therefore the reported spectrum is made directly from all successfully escaped packets and has no viewing-cone selection. Because the problem is spherical and static, this is the natural angle-integrated comparison with the analytical solution.

Core skipping is enabled with `xcritmax=3`. This accelerates the very large number of nearly local line-core scatterings and makes the high-optical- depth experiments practical. It is an explicitly documented approximation. It should preserve the main escape spectrum when used appropriately, but at the lowest $a_v\tau_0$ it can be one contributor to small trough, width, or wing differences. I do not treat core skipping as a proven explanation for every residual; the main known limitation at $\tau_0 = 10^5$ is the asymptotic nature of the analytical expression itself.

10 Reading the RASCAS result binary

The escaped result `Cont.res` is also an unformatted sequential Fortran file. I read the following records in their written order:

1. total packet count;
2. packet IDs;
3. terminal status flags;
4. final positions;
5. external-frame frequencies ν;
6. final propagation directions;
7. scattering counts;
8. propagation times.

Only `status=1` packets enter the escaped spectrum. I preserve and report all status counts, so lost or unfinished packets cannot disappear silently. Every packet in every completed comparison reported here has status 1.

11 Analytical static-sphere spectrum

11.1 Dimensionless frequency

I convert each escaped external-frame frequency to Doppler units:

$$x = \frac{\nu - \nu_\alpha}{\Delta\nu_D}, \qquad \nu_\alpha = \frac{c}{\lambda_\alpha}. \tag{15}$$

With this frequency convention, $x < 0$ is red and $x > 0$ is blue. A wavelength offset would have the opposite sign near line centre, so I never mix the two conventions.

11.2 Analytical equation and normalization

For a homogeneous static sphere I evaluate

$$J(x) = \sqrt{\frac{\pi}{6}} \frac{1}{4a_{\rm v}\tau_0} \frac{x^2}{1 + \cosh\left[\sqrt{\frac{2\pi^3}{27}} \frac{|x|^3}{a_{\rm v}\tau_0}\right]}. \tag{16}$$

The convention in 16 satisfies

$$\int_{-\infty}^{\infty} J(x)\, dx = \frac{1}{4\pi}.$$

My Monte Carlo histogram is a unit-area probability density. The directly comparable analytical curve is consequently

$$p_{\rm analytic}(x) = 4\pi J(x), \qquad \int p_{\rm analytic}(x)\, dx = 1. \tag{17}$$

I evaluate the full curve at the histogram-bin centres. I normalize its sampled finite grid only for negligible quadrature truncation; I never change its width, optical depth, or amplitude to fit RASCAS.

The approximate peak position is

$$|x_{\rm peak}| \simeq 0.931(a_{\rm v}\tau_0)^{1/3}. \tag{18}$$

Thus increasing optical depth forces photons farther into the wings before escape and increases the peak separation approximately as $\tau_0^{1/3}$.

12 Constructing the numerical spectrum and uncertainties

I histogram the escaped x values over $-100 \le x \le 100$ with bin width $\Delta x = 0.5$. If bin i contains N_i packets and the total escaped count is $N_{\rm esc}$, I calculate

$$p_{{\rm MC},i} = \frac{N_i}{N_{\rm esc}\Delta x}, \qquad \sigma_i = \frac{\sqrt{N_i}}{N_{\rm esc}\Delta x}. \tag{19}$$

Therefore

$$\sum_i p_{{\rm MC},i}\Delta x = 1.$$

The raw count is always retained. A smooth normalized curve alone can hide whether a tail bin contains ten photons or ten thousand photons.

My main full-profile statistic is

$$L_1 = \sum_i |p_{\mathrm{MC},i} - p_{\mathrm{analytic},i}|\Delta x. \tag{20}$$

I also measure peak positions, peak separation, integrated red/blue packet ratio, symmetry fraction

$$S = \frac{|N_{\mathrm{red}} - N_{\mathrm{blue}}|}{N_{\mathrm{red}} + N_{\mathrm{blue}}},$$

central probability $P(|x| < 1)$, RMS density difference, mean scattering count, and a chi-square-like diagnostic. The latter is not interpreted as a formal p-value because the normalized bins are correlated and the analytical diffusion solution is approximate.

13 Meaning of every generated plot

Linear spectrum overlay This is the most direct physical comparison. It emphasizes peak positions, peak heights, widths, the central trough, and the probability-carrying portion of the spectrum. Agreement here is more important than visual agreement at probabilities many orders of magnitude below the peak.

Logarithmic spectrum overlay This displays the weak wings and rare events. It is valuable for finding an unexpected long tail, but it visually magnifies tiny absolute differences. Bins with zero packets are plotted at a small positive floor so they can appear on logarithmic axes; vertical drops to that floor are plotting artifacts, not negative probability. Isolated high tail bins can contain only one or a few photons.

Folded red and blue halves I reflect the red half about $x = 0$ and compare it with the blue half versus $|x|$. This removes the common double-peaked shape and directly exposes directional or frequency-sign asymmetry. A static sphere should agree within Monte Carlo noise.

Absolute residual This shows $p_{\mathrm{MC}} - p_{\mathrm{analytic}}$. Positive residuals identify where RASCAS assigns more probability, and negative residuals where it assigns less. It reveals whether a lower peak is compensated by broader shoulders or wings.

Fractional residual This shows $(p_{\mathrm{MC}} - p_{\mathrm{analytic}})/p_{\mathrm{analytic}}$ only where the analytical curve exceeds 10^{-4} of its maximum. Without this mask, division by a vanishing analytical tail would create huge but meaningless ratios.

Poisson count plot This displays N_i with $\sqrt{N_i}$ error bars. It is the clearest view of the actual Monte Carlo information content and distinguishes statistically meaningful discrepancies from sparsely populated tail fluctuations.

14 Physical origin of the double-peaked spectrum

A photon emitted at line centre sees the largest HI cross-section and therefore has a very short mean free path. Repeated resonant scatterings produce coupled random walks in position and frequency. Thermal atomic velocities move a photon toward either frequency wing. In a wing the cross-section is smaller, the mean free path is longer, and a sufficiently large frequency excursion can carry the photon to the boundary. Because the gas is static and symmetric, red and blue excursions are equally likely.

The expected signatures are consequently a suppressed line centre, two peaks, equal integrated red and blue flux, and increasing peak separation with $(a_v\tau_0)^{1/3}$. The simulation reproduces every one of these signatures.

15 Quantitative results

Table 2: Angle-integrated RASCAS comparisons. The first three rows use 10^5 packets; the fourth is the $\tau_0 = 10^5$ photon-count refinement.

τ_0	N_γ	$a_v\tau_0$	analytic $\lvert x_p\rvert$	RASCAS $x_{p,r}$	RASCAS $x_{p,b}$	separation	red/blue	S	L_1
10^5	10^5	471.873	7.2481	−7.25	6.75	14.0	1.00977	0.00486	0.11329
10^6	10^5	4718.730	15.6156	−15.25	15.25	30.5	1.00337	0.00168	0.06492
10^7	10^5	47187.295	33.6427	−32.75	33.25	66.0	1.01219	0.00606	0.04462
10^5	10^6	471.873	7.2481	−7.25	7.25	14.5	0.99936	0.000322	0.11243

All rows have unit Monte Carlo and analytical normalization, and all requested packets escape with status 1. The systematic improvement in L_1 with optical depth is important: it is the behavior expected if the remaining mismatch is mainly the finite-$a_v\tau_0$ limitation of an asymptotic analytical solution, not a hidden unit error.

15.1 $\tau_0 = 10^5$: diagnostic case

Here $a_v\tau_0 = 471.873$, below the conservative value of roughly 10^3 often used to identify the strong wing-diffusion regime. With 10^5 photons, the measured peaks are -7.25 and $+6.75$; the half-bin displacement of the blue maximum is compatible with $\Delta x = 0.5$ binning and sampling noise. The red/blue ratio is 1.00977 and $L_1 = 0.11329$.

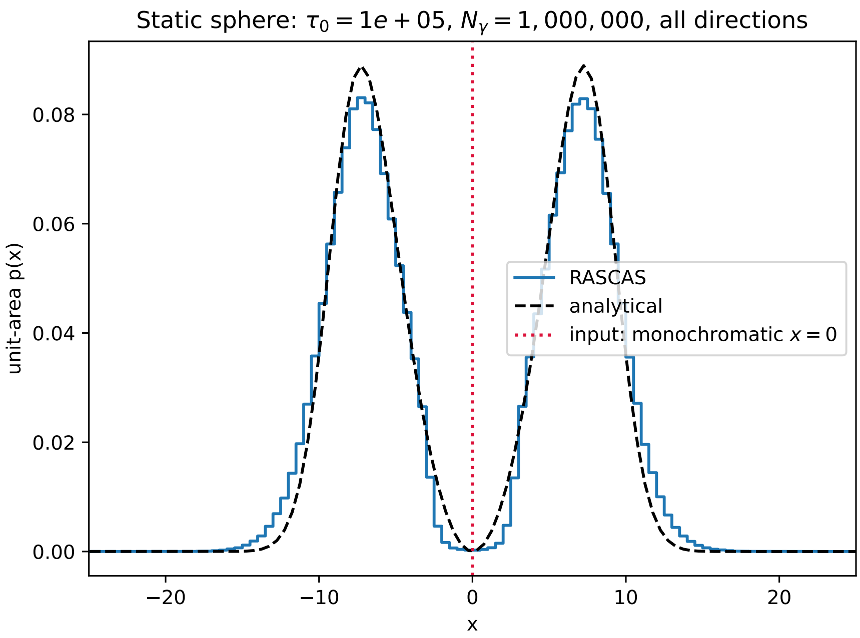
With 10^6 photons, both maxima lie at ± 7.25, essentially exactly matching the analytical $|x_p| = 7.2481$. The red/blue ratio improves to 0.999356 and the symmetry fraction to 3.22×10^{-4}. However, L_1 changes only slightly, from 0.11329 to 0.11243. This is decisive: increasing the sample by a factor of ten removes visible noise and improves symmetry, but does not remove the broader RASCAS shoulders and wings. The remaining difference is systematic rather than a shortage of photons.

The one-million-packet peak density is approximately 0.0831, compared with 0.08894 analytically, about 6.7% lower. Since both curves have unit area, the lower numerical peak is compensated by broader shoulders and heavier tails. For example, 48 of the million packets occur at $|x| \geq 18$. These events have only 4.8×10^{-5} total probability, yet they appear prominent on a log axis. This explains why the logarithmic plot can look substantially different even when the peak positions and the probability-dominating part of the profile agree well.

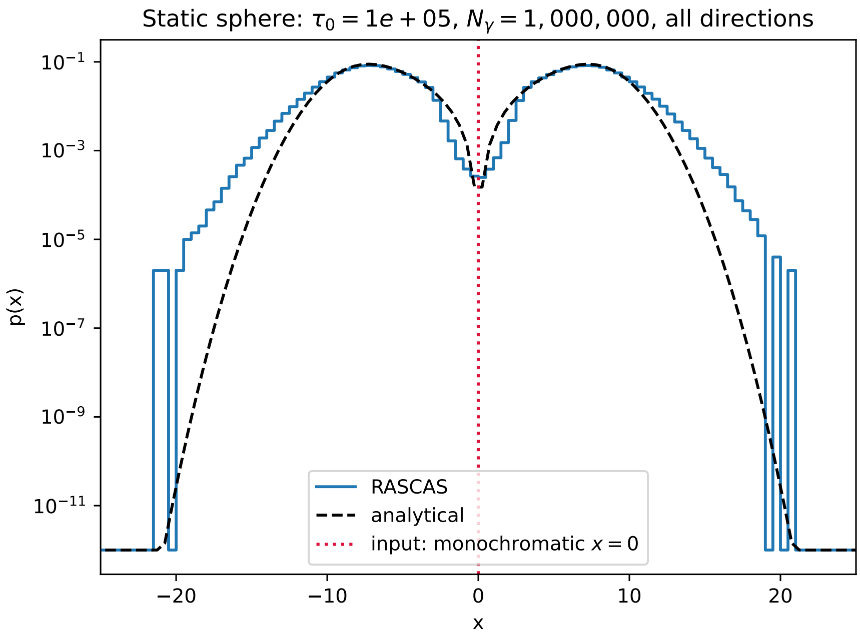
The correct conclusion is therefore "pass as a lower-$a_v\tau_0$ diagnostic," not "exact asymptotic-profile match." The main explanation is that 16 is a diffusion/wing approximation and $a_v\tau_0 \simeq 472$ is marginal. Finite bin width, the finite 64^3 mesh, and the documented core-skipping acceleration can contribute smaller numerical effects. None of these produces a nonphysical asymmetry or an optical-depth shift.

15.2 $\tau_0 = 10^6$: strong analytical validation

At $\tau_0 = 10^6$, $a_v\tau_0 = 4718.73$, so the system is well inside the high-optical-depth regime. The analytical peak magnitude is $|x_p| = 15.6156$. The RASCAS peaks occur at -15.25 and $+15.25$, within one $\Delta x = 0.5$ bin of the prediction. The peak separation is 30.5 compared with the approximate analytical value $2|x_p| = 31.2312$.



(a) Linear scale: peak position and overall probability-carrying shape.



(b) Log scale: rare wings are magnified; isolated tail steps contain very few photons.

Figure 1: $\tau_0 = 10^5$ with one million photons.

The integrated red/blue ratio is 1.00337 and the symmetry fraction is 0.00168. Only 5×10^{-5} of packets remain within $|x| < 1$, demonstrating the expected strong line-centre suppression. The numerical maximum differs from the sampled analytical maximum by about -1.95%, and the full-profile $L_1 = 0.06492$ is comfortably below the validation limit of 0.15. This is a strong pass: location, height, symmetry, normalization, and full shape all agree.

15.3 $\tau_0 = 10^7$: strongest completed asymptotic test

For $\tau_0 = 10^7$, $a_v\tau_0 = 47187.3$. The analytical peak position is $|x_p| = 33.6427$, while RASCAS gives -32.75 and $+33.25$. The numerical separation is 66.0 compared with the approximate analytical value 67.2855. This difference is small relative to the peak frequency and includes the finite histogram-bin resolution.

All 100,000 packets escape. The red/blue ratio is 1.01219, corresponding to a symmetry fraction of 0.00606, statistically reasonable for two halves of a 100,000-packet sample. No photon lies within $|x| < 1$. The numerical peak is about 4.2% above the sampled analytical maximum, and $L_1 = 0.04462$, the best of the three optical-depth cases. This monotonic improvement from 10^5 to 10^7 is strong evidence that the physical units and optical-depth convention are correct.

16 Monte Carlo noise, binning, and systematic differences

For independent counts, relative Poisson noise scales approximately as $N_i^{-1/2}$ and global Monte Carlo fluctuations scale as $N_\gamma^{-1/2}$. Increasing the total packet count from 10^5 to 10^6 should reduce random noise by roughly $\sqrt{10} \simeq 3.16$. The $\tau_0 = 10^5$ convergence run does exactly what this predicts: red–blue symmetry becomes much tighter and the histogram becomes smoother.

Noise reduction does not guarantee convergence to the plotted analytical curve, because the analytical curve is not an exact numerical solution for arbitrary $a_v\tau_0$. The nearly unchanged L_1 at $\tau_0 = 10^5$ demonstrates a systematic difference between the simulation and the asymptotic expression. This is physically different from Monte Carlo noise.

The bin width $\Delta x = 0.5$ limits peak localization to approximately a bin. A reported peak at 6.75 rather than 7.25 in a 100,000-packet histogram does not mean the physical peak shifted by a precisely measured 0.5; neighboring bins near the smooth maximum can exchange rank because of sampling fluctuations. Raw counts and Poisson errors must be inspected before assigning significance.

The far-wing discrepancy looks largest on logarithmic plots. However, a difference between 10^{-8} and 10^{-5} is visually three decades while contributing very little to unit-area L_1. Conversely, a small vertical difference across the broad peak can carry much more probability. I therefore use the linear plot, raw counts, L_1, peak positions, and symmetry together, rather than judging correctness from a log plot alone.

17 Failure modes explicitly ruled out

The following possible mistakes are ruled out by direct measurements:

- **Wrong converter density scaling:** the post-conversion density is exactly 232.31332352 times the Cartesian input and yields the requested $n_{\rm HI}$.
- **Radius/diameter confusion:** rays integrate from the centre to the exact radius $R = 0.3$, and measured τ_0 equals the requested value.
- **Wrong temperature:** both RAMSES and domain readers recover 100 K.

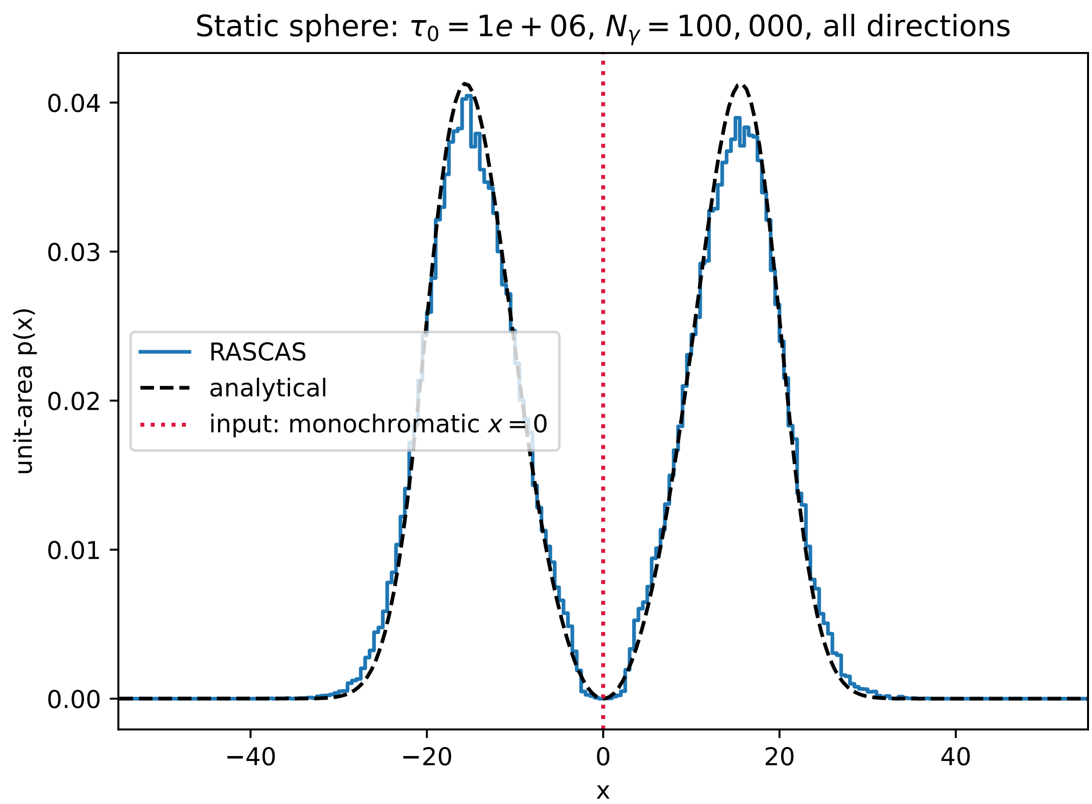


Figure 2: $\tau_0 = 10^6$, 100,000 photons. This is a stronger analytical test because $a_v\tau_0 \gg 10^3$.

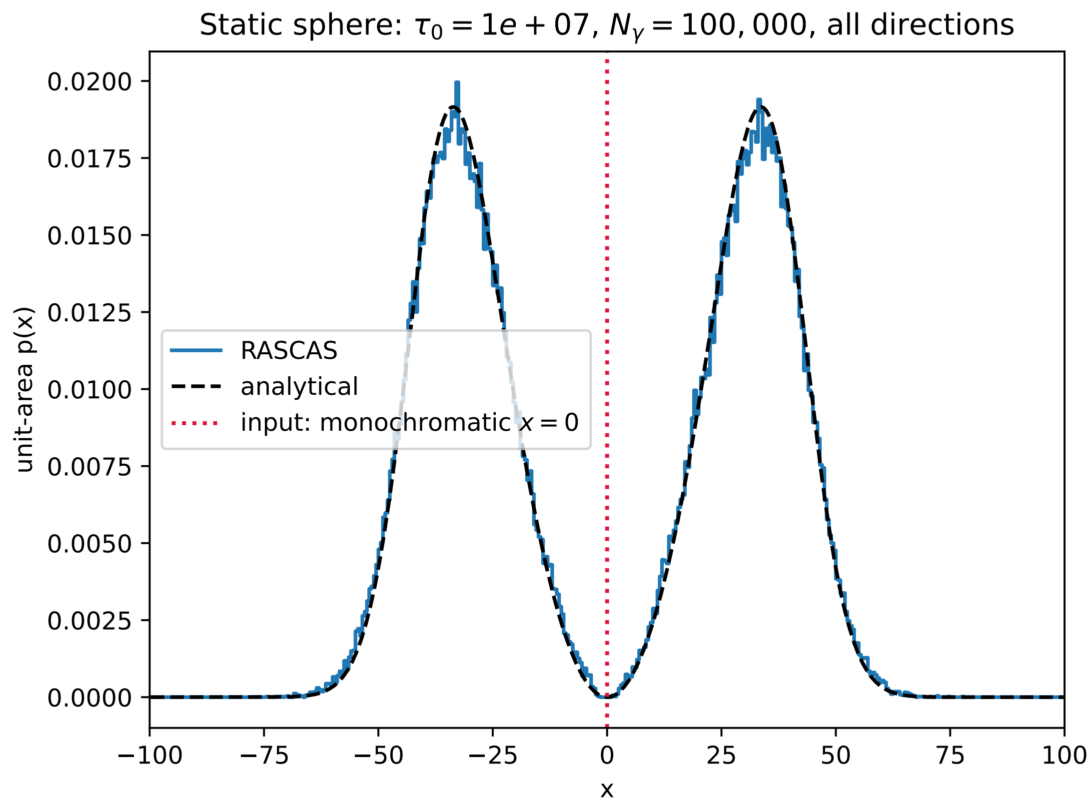


Figure 3: $\tau_0 = 10^7$, 100,000 photons. This is the strongest completed comparison with the asymptotic analytical solution.

- **Wrong ionization convention:** RAMSES `scalar_01=xHII=0` is verified upstream; RASCAS recovers the expected neutral hydrogen density.
- **Wrong field order:** temperature, metallicity, velocity, and HI density are independently checked after reading the generated binaries.
- **Velocity or geometrical asymmetry:** all input velocities are zero, the boundary is a centred sphere, and folded red/blue spectra agree statistically.
- **Missing packets:** every completed result has the requested count and every packet has successful terminal status 1.
- **Normalization error:** both numerical and analytical spectra integrate to unity.
- **Frequency-sign error:** frequency, not wavelength, defines x; negative x is consistently labeled red.

18 Overall assessment

The Cartesian-to-RAMSES-to-RASCAS interface is validated. The unit conversion, temperature, neutral fraction, field order, octree leaf count, spherical boundary, and optical depth all pass independent checks. The transport produces the expected symmetric double peak and the expected growth of peak separation with optical depth.

The $\tau_0 = 10^5$ case passes as a diagnostic with finite-diffusion-limit and documented acceleration caveats. Its one-million-photon run proves that the remaining broader wings are not simply low photon statistics. The $\tau_0 = 10^6$ and 10^7 cases are strong analytical passes, with peak-height differences of about 2% and 4%, respectively, and L_1 values of 0.0649 and 0.0446. The improvement with $a_{\rm v}\tau_0$ is physically sensible and supports the conclusion that no hidden unit or optical-depth rescaling is required.

A Reproducibility map

The principal machine-readable and plotting files are:

- `analysis/physics.py`: constants and all physical conversions;
- `cartesian_input/generate_cartesian.py`: array generation and reload checks;
- `analysis/validate_photons.py`: streaming Fortran IC validation;
- `analysis/compare_spectrum.py`: escaped-result decoding, histogram, analytical curve, metrics, CSV, and plots;
- `analysis/<case>/ramses/validation.json`: converted-field checks;
- `analysis/<case>/domain/validation.json`: spherical-domain and ray checks;
- `analysis/<case>/<run>/spectrum_metrics.json`: numerical comparison;
- `analysis/<case>/<run>/spectrum.csv`: every spectral bin, raw count, Poisson error, analytical value, and residual;
- `CODEX_WORKLOG.md`: exact execution history, scheduler jobs, failures, diagnoses, and results.

B Compact calculation recipe

For another target optical depth at the same temperature and radius, I would:

1. calculate v_{th}, $\Delta\nu_D$, a_{v}, and σ_0 with the installed RASCAS constants;
2. calculate $n_{\text{HI}} = \tau_0/(\sigma_0 R)$;
3. calculate ρ_{RAMSES} from 10;
4. divide by 232.31332352 before writing the Cartesian density array;
5. calculate u from 8 and write the remaining homogeneous fields;
6. convert to RAMSES and verify all 262,144 leaves and field values;
7. create the radius-0.3 domain and measure τ_0 with clipped rays;
8. generate and validate line-centre photons;
9. run RASCAS and require a valid status for every packet;
10. compute x with 15, construct 19, evaluate 16, and compare using 20 and the diagnostic plots.